

Predicting clogging from the contact network: a graph neural network for deformable-particle suspensions at a microfluidic constriction

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Abstract

When a suspension of soft particles is driven through a narrow gap, it either flows or it jams: a load-bearing arch can form across the constriction and arrest the flow. This clogging transition governs everything from coffee filters and household drains to grain hoppers and the obstruction of small blood vessels, yet most of what we know about it comes from rigid particles and from after-the-fact description rather than prediction. Here we ask a different question: given a single snapshot of which particles are touching which, can one predict whether the suspension will clog — without integrating the dynamics forward? We address it with a two-dimensional lattice-Boltzmann immersed-boundary model of deformable capsules pushed through a rectangular neck, run across ten aperture ratios spanning the flowing and clogged regimes. Each simulation frame is converted into a graph in which capsules are nodes and near-contacts are edges carrying the local gap and approach rate, and an edge-aware graph neural network is trained to classify the configuration as clogging or flowing. The simulations reproduce the expected arching threshold near an aperture-to-particle ratio $D/d \approx 3$. The classifier reaches an area under the ROC curve of 0.93 on held-out frames, and — under a leave-one-run-out test that removes all correlation between training and test configurations — it generalises to runs it has never seen with a mean accuracy of 0.87, failing only at the single aperture sitting exactly on the transition, where the outcome is genuinely ambiguous. The result shows that the instantaneous contact network carries a learnable, transferable signature of imminent clogging in deformable suspensions, and it provides a reproducible pipeline from mesoscale simulation to graph-based prediction.

Keywords: clogging · jamming · soft-particle suspensions · contact network · graph neural network · lattice Boltzmann method · immersed boundary method

1 Introduction

Push a crowd of particles through an opening only a few particle diameters wide and one of two things happens. Either the particles file through and the flow continues, or a handful of them lock into a stable arch that spans the opening and brings the flow to a halt. This clogging, or jamming at a bottleneck, is one of the most familiar failure modes in nature and technology: it is why coffee grounds bridge a filter, why a kitchen drain blocks with debris, why grain stalls in a silo, and why circulating cells and particulates can obstruct a narrow vessel. It is also a long-standing problem in the statistical physics of disordered and athermal systems, because the arch is a collective,

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force-bearing structure that forms intermittently and is sensitive to the microscopic arrangement of contacts [2, 11].

A large body of experimental and computational work has mapped when clogging occurs. For rigid grains and hard spheres flowing through a constriction, the controlling quantity is the ratio of the aperture width to the particle diameter: permanent clogs form below an aperture-to-particle ratio of roughly two to four, where a small number of particles can bridge the gap, while wider openings sustain continuous or intermittent flow [12, 15]. Softness changes the picture. Deformable particles can squeeze through gaps that would arrest rigid ones, so the clogging boundary shifts and the dynamics of the arch differ from the granular case [1, 6]. Most of this literature, however, is descriptive: it characterises the statistics of clogging events, the arch lifetimes, and the dependence on geometry and volume fraction. Comparatively little of it is *predictive* at the level of an individual configuration, and almost none of it treats the clog as a property of the contact network that could be read off directly.

Machine learning on graphs offers a natural language for that question. A suspension at an instant in time is, after all, a graph: the particles are nodes and their mechanical contacts are edges. Graph neural networks (GNNs) learn functions on exactly this kind of irregular, relational data by passing messages along edges [3, 5, 7], and they have recently been used as fast surrogates for the *dynamics* of particulate suspensions, for example by learning the hydrodynamic mobility of a configuration so that its motion can be integrated cheaply [10]. Those efforts target rigid particles and the forward problem of time evolution. The complementary question — whether a GNN can read the contact network of a *deformable* suspension and predict a collective outcome such as clogging, without any time integration at all — has not, to our knowledge, been addressed.

That is the question of this paper. We build a two-dimensional lattice-Boltzmann immersed-boundary (LBM-IBM) model of soft capsules driven through a rectangular constriction, run it across a range of aperture ratios that brackets the flowing and clogged regimes, and from the resulting trajectories extract, for every saved frame, the contact graph of the suspension. We then train an edge-aware GNN to classify each graph as belonging to a clogging or a flowing run, and we evaluate it honestly: not only on randomly held-out frames, but with a leave-one-run-out protocol in which an entire aperture is removed from training, so that the test configurations are statistically independent of those the network has seen. The simulations reproduce the expected arching threshold near $D/d \approx 3$; the classifier attains an area under the ROC curve of 0.93 on held-out frames and a leave-one-run-out accuracy of 0.87, with its only weakness lying exactly at the aperture on the transition. The contribution is twofold: a demonstration that the instantaneous contact network carries a learnable and transferable signature of clogging in soft suspensions, and a reproducible pipeline that connects a mesoscale physics solver to a graph-based predictor.

The remainder of the paper is organised as follows. Section 2 sets out the fluid solver, the capsule and contact models, the geometry and driving, the clogging diagnostic, the graph construction, and the network and its training. Section 3 reports the clogging transition, the classifier performance, the leave-one-run-out generalisation, and the structural difference between the regimes. Section 4 interprets the findings, reports a negative result on data augmentation, and states the limitations. Section 5 concludes.

2 Model and methods

The simulations use an in-house C++ solver that couples a lattice-Boltzmann fluid to immersed deformable capsules; all quantities below are in lattice units (LU). Sections 2.1–2.3 describe the physics, Section 2.4 the clogging diagnostic, and Sections 2.5–2.7 the graph representation and the neural network.

2.1 Lattice-Boltzmann fluid

The fluid is solved on a regular two-dimensional grid with the D2Q9 velocity set $\{\mathbf{e}_i\}_{i=0}^8$, weights $w_0 = 4/9$, $w_{1-4} = 1/9$, $w_{5-8} = 1/36$, and lattice sound speed $c_s^2 = 1/3$ [8]. Let $f_i(\mathbf{x}, t)$ be the discrete distribution in direction i . The density and momentum are

$$\rho = \sum_i f_i, \quad \rho \mathbf{u} = \sum_i f_i \mathbf{e}_i + \frac{1}{2} \mathbf{F} \Delta t, \quad (1)$$

and the equilibrium is the second-order Hermite truncation of the Maxwellian,

$$f_i^{\text{eq}} = w_i \rho \left[1 + \frac{\mathbf{e}_i \cdot \mathbf{u}}{c_s^2} + \frac{(\mathbf{e}_i \cdot \mathbf{u})^2}{2c_s^4} - \frac{\mathbf{u} \cdot \mathbf{u}}{2c_s^2} \right]. \quad (2)$$

We evolve f_i by a single-relaxation-time (BGK) collide-and-stream step with a regularisation of the non-equilibrium part to suppress ghost modes [9],

$$f_i(\mathbf{x} + \mathbf{e}_i \Delta t, t + \Delta t) = f_i(\mathbf{x}, t) - \frac{1}{\tau} [f_i - f_i^{\text{eq}}] + F_i \Delta t, \quad (3)$$

with relaxation time τ fixing the kinematic viscosity through $\nu = c_s^2(\tau - \frac{1}{2})\Delta t$. We use $\tau = 0.9$, i.e. $\nu = 0.133$ LU. A body force $\mathbf{F} = (F_x, 0)$ drives the flow and enters the collision through the second-order forcing of Guo et al. [4],

$$F_i = \left(1 - \frac{1}{2\tau}\right) w_i \left[\frac{\mathbf{e}_i - \mathbf{u}}{c_s^2} + \frac{(\mathbf{e}_i \cdot \mathbf{u}) \mathbf{e}_i}{c_s^4} \right] \cdot \mathbf{F}. \quad (4)$$

No-slip walls at the top and bottom of the channel are imposed by halfway bounce-back; the streamwise direction is periodic, so the suspension recirculates and maintains its concentration. The channel is $N_x \times N_y = 220 \times 48$ LU. The driving $F_x = 2.24 \times 10^{-5}$ was calibrated to give a developed centre-line speed $u_{\text{max}} \approx 8 \times 10^{-3}$ LU, so that the Reynolds number $\text{Re} = u_{\text{max}} r / \nu \approx 0.24$ and the Mach number $u_{\text{max}} / c_s \approx 0.014$ both sit well inside the incompressible, Stokes-like regime appropriate to microfluidic flow.

2.2 Immersed deformable capsules

Each capsule is a closed loop of $N_n = 16$ Lagrangian nodes $\mathbf{X}_\ell$ enclosing fluid, with rest diameter $d = 8$ LU (radius $r = 4$). Two-way coupling to the fluid uses the immersed-boundary method [13]: membrane forces are spread to the lattice and the node velocities are interpolated back, both with the four-point Peskin regularised delta δ_h ,

$$\mathbf{f}(\mathbf{x}) = \sum_\ell \mathbf{F}_\ell \delta_h(\mathbf{x} - \mathbf{X}_\ell) \Delta s_\ell, \quad \dot{\mathbf{X}}_\ell = \sum_{\mathbf{x}} \mathbf{u}(\mathbf{x}) \delta_h(\mathbf{x} - \mathbf{X}_\ell) \Delta x^2. \quad (5)$$

The spreading-interpolation cycle is iterated $K = 2$ times per step (multi-direct forcing) to reduce the residual slip at the membrane. The membrane resists shear and area change through a Skalak constitutive law [14] with shear modulus G_s and area modulus $C = 10 G_s$, plus discrete bending ($k_b = 0.02$), area ($k_a = 1.0$) and perimeter ($k_p = 0.1$) penalties. The capsule deformability is set by the capillary number $\text{Ca} = \rho \nu r \dot{\gamma} / G_s$ with the wall shear rate $\dot{\gamma} = 4u_{\text{max}} / N_y$; we fix $\text{Ca} = 0.01$ for all runs, which fixes G_s given the flow.

Two capsules (or a capsule and a wall or obstacle) interact through a short-range repulsion that prevents membrane interpenetration. Between nodes a and b at separation r_{ab} below a cut-off r_c the force along the unit normal $\hat{\mathbf{n}}_{ab}$ is

$$\mathbf{F}_{ab}^{\text{rep}} = \epsilon \left(\frac{\sigma}{r_{ab}} \right)^p \hat{\mathbf{n}}_{ab}, \quad r_{ab} < r_c, \quad (6)$$

with $\epsilon = 10^{-3}$, $\sigma = 1$, $p = 4$, $r_c = 2$ LU. Near-contact hydrodynamics below the grid resolution is captured by a lubrication correction applied when the surface gap falls below 1.5 LU. (The solver also supports an optional dissipative-frictional contact, but it is disabled here so that the contacts are governed by the conservative repulsion alone.)

2.3 Geometry, seeding and driving

The constriction is a pair of rectangular obstacles spanning $x \in [150, 175]$, one growing from each wall, leaving a centred gap of width $D = (D/d)d$. The single control parameter is the aperture-to-particle ratio D/d . Upstream of the neck, a dense plug of capsules is seeded in the region $x \in [40, 145]$, $y \in [8, 40]$ on a hexagonal grid whose spacing $2r + \delta_{\min}$ guarantees a minimum surface gap $\delta_{\min} = 1$ LU by construction; this deterministic “fill” seeding reliably packs ≈ 40 capsules, where rejection-based random seeding leaves the region sparse. The body force then drives the plug toward and through the neck, and because the channel is periodic the capsules recirculate, so the system sustains a roughly constant concentration at the constriction rather than draining.

Each run is initialised, ramped over a 2000-step warm-up, and then integrated for 10^4 production steps with the capsule trajectory written every 200 steps. We sweep ten apertures, $D/d \in \{1.5, 2.0, 2.5, 2.75, 3.0, 3.25, 3.5, 4.0, 4.5, 5.0\}$, at fixed $\text{Ca} = 0.01$, spanning the clogged and flowing regimes with extra sampling near the transition.

2.4 Clogging diagnostic

For each capsule we record the first time its centroid crosses the neck centre $x_{\text{neck}} = 162.5$, and define the escaped fraction $\eta(t)$ as the cumulative number of capsules that have crossed, divided by the total. A run is labelled *clogged* when, at the end of the run, $\eta < 0.9$ and the escape flux has stalled (no new crossing over the final window) while capsules remain upstream; otherwise it is *flowing*. As a structural diagnostic we also compute the neck contact number Z , the mean number of contacting neighbours per capsule within a band $|x - x_{\text{neck}}| \leq 15$ LU, where a contact is any pair with surface gap below 1 LU.

2.5 Graph representation

Each saved frame is turned into an undirected graph $\mathcal{G} = (V, E)$. The nodes V are the capsules; the edges E connect every pair in near-contact, i.e. capsules i, j with

$$\|\mathbf{c}_i - \mathbf{c}_j\| \leq r_i + r_j + \delta_c, \quad \delta_c = 1 \text{ LU}, \quad (7)$$

where $\mathbf{c}_i$ is the centroid; the streamwise minimum image is used because the channel is periodic. Each node carries a nine-dimensional feature vector — normalised position $(x/N_x, y/N_y)$, velocity (u_x, u_y) , speed, normalised distance to the nearer wall, type, radius, and contact degree — and each (directed) edge carries six features: the centre distance, the surface gap $g_{ij} = \|\mathbf{c}_i - \mathbf{c}_j\| - (r_i + r_j)$, the unit separation vector, the relative speed, and the approach rate $-(\mathbf{v}_i - \mathbf{v}_j) \cdot \hat{\mathbf{n}}_{ij}$ (positive when the pair is closing). The gap and approach rate are the features that most directly encode an incipient arch, so the network is given the means to use them. Each graph is labelled by the clog/flow outcome of the run it belongs to.

2.6 Graph neural network

We classify a graph with an edge-conditioned message-passing network [3]. Node features are first embedded, $\mathbf{h}_i^{(0)} = W_{\text{in}} \mathbf{x}_i$, into a hidden width $H = 48$. Each of $L = 3$ message-passing layers updates the node states using both neighbour states and the edge features $\mathbf{e}_{ij}$:

$$\mathbf{m}_i^{(\ell)} = \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} \phi([\mathbf{h}_i^{(\ell)} \parallel \mathbf{h}_j^{(\ell)} \parallel \mathbf{e}_{ij}]), \quad (8)$$

$$\mathbf{h}_i^{(\ell+1)} = \mathbf{h}_i^{(\ell)} + \text{LN}\left(\psi([\mathbf{h}_i^{(\ell)} \parallel \mathbf{m}_i^{(\ell)}])\right), \quad (9)$$

where ϕ and ψ are two-layer perceptrons with ReLU activations, $[\cdot \parallel \cdot]$ is concatenation, LN is layer normalisation and the residual connection stabilises the depth. After the message-passing layers the node states are pooled to a single graph vector by concatenating their mean and maximum,

$$\mathbf{g} = [\frac{1}{|V|} \sum_i \mathbf{h}_i^{(L)} \parallel \max_i \mathbf{h}_i^{(L)}], \quad (10)$$

and a final perceptron maps $\mathbf{g}$ to a single logit z whose sigmoid $\hat{p} = \sigma(z)$ is the predicted clogging probability. The model has $\approx 1.5 \times 10^4$ parameters.

2.7 Training and evaluation

The network is trained with the class-weighted binary cross-entropy

$$\mathcal{L} = -\frac{1}{B} \sum_{k=1}^B [w_+ y_k \log \hat{p}_k + (1 - y_k) \log(1 - \hat{p}_k)], \quad w_+ = \frac{n_-}{n_+}, \quad (11)$$

where $y_k \in \{0, 1\}$ is the clog label and $w_+ = n_-/n_+$ up-weights the minority class so the optimiser cannot collapse onto the majority. We use the Adam optimiser (learning rate 3×10^{-3} , weight decay 10^{-4}) for 120 epochs with batches of 16 graphs.

Two evaluations are reported. The first is a conventional random split: the pooled frames are divided 75/25 into training and validation, and we report the validation accuracy, the balanced accuracy (mean of the two per-class recalls, which is the meaningful figure under class imbalance), and the area under the ROC curve. The second, and the one we emphasise, is *leave-one-run-out* (LORO): each aperture is removed in turn, the network is retrained on the remaining apertures, and we measure the fraction of the held-out run’s frames that are classified correctly. Because frames from the same simulation are strongly correlated, the random split leaks information between training and validation; the LORO protocol removes that leak and tests genuine generalisation to configurations the network has never encountered.

3 Results

The data set comprises ten runs — one per aperture — and 296 contact graphs (one per saved frame), of which 120 belong to clogging runs and 176 to flowing runs.

3.1 The clogging transition

Figure 1 shows the two regimes directly. In a narrow neck (top row) the dense plug advances, reaches the constriction, and arrests: the capsules at the throat lock into an arch and the whole plug slows to a halt (the capsules turn dark, i.e. nearly stationary), even though the fluid continues to squirt through the gap. In a wider neck (bottom row) the same plug accelerates as it approaches the constriction and the capsules stream through it (they turn red, i.e. fast), so the plug empties downstream. The contrast between the final frames — a stalled, arrested plug versus a fast, draining one — is the macroscopic signature of clogging that the rest of the paper quantifies and then learns to predict.

Figure 2 shows the two run-level diagnostics as functions of the aperture ratio. The neck contact number Z rises smoothly with D/d , from about one contact per capsule at the tightest necks to more than two at the widest, simply because wider gaps admit denser local packing in the throat band. The clog label, by contrast, switches sharply: every run with $D/d \leq 2.75$ clogs, and every run with $D/d \geq 3.0$ flows. The transition therefore sits between $D/d = 2.75$ and 3.0 , in agreement with the arching threshold of $D/d \approx 2-3$ established for constricted suspensions of both rigid and soft particles [1, 6, 12]. The escaped fraction increases monotonically with aperture across the flowing runs, confirming that the label reflects a genuine change in transport and not an artefact of the threshold.

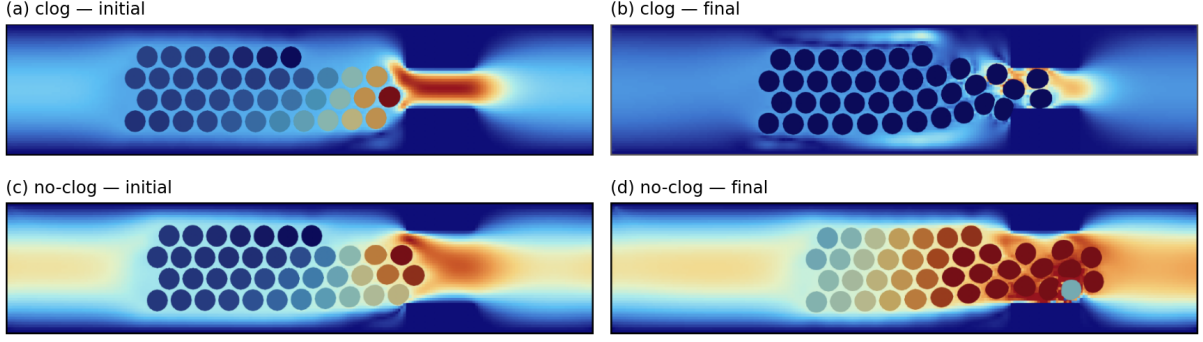


Figure 1. Representative LBM-IBM frames for a clogging (top, narrow neck) and a flowing (bottom, wide neck) run, at the start (left) and end (right) of the simulation. The background is the fluid speed field and each capsule is coloured by its own speed (blue slow, red fast); flow is left to right. In the clogging run the capsules arrest at the constriction and the plug remains dark (slow); in the flowing run they accelerate (red) and pass through.

3.2 Classifying configurations

Trained on the pooled frames, the network learns to separate clogging from flowing configurations well. On the held-out frames it reaches an accuracy of 0.78, a balanced accuracy of 0.78, and an area under the ROC curve of 0.93 (Figure 3). The ROC curve lies well above the diagonal across the whole range of operating points, so the model ranks clogging configurations above flowing ones reliably; the class-weighted loss prevents it from defaulting to the majority class, which is why the balanced accuracy tracks the raw accuracy despite the 120:176 class imbalance.

3.3 Generalisation across runs

The leave-one-run-out test is the more demanding and more informative one. Figure 4 reports the per-aperture accuracy when each run is removed from training in turn. The mean over the ten folds is 0.87: the network classifies the frames of an unseen aperture correctly 87% of the time on average, with 0.92 on the flowing runs and 0.79 on the clogging runs. The accuracy is high and stable away from the transition — 0.97–1.00 at the clear-clog apertures $D/d \leq 2.0$ and the clear-flow apertures $D/d \geq 4.0$ — and collapses at exactly one fold, $D/d = 2.75$, where it reaches only 0.43. That fold is the last clogging aperture before the transition; when it is removed, the training set contains clogging examples no closer than $D/d = 2.5$ and flowing examples no closer than $D/d = 3.0$, so the network has seen nothing at the boundary and is forced to extrapolate across it. The failure is therefore localised exactly where the physical outcome is itself most ambiguous, rather than being a systematic weakness of the model.

3.4 Where the contacts concentrate

The two regimes are distinguished not by the overall density of the contact network — both are dense suspensions — but by where in the channel the contacts sit relative to the neck. Figure 5 averages each capsule’s contact degree over its streamwise position, pooled separately over all frames of the clogging and the flowing runs. Upstream of the neck the two ensembles are similar, as expected for a dense plug. At the constriction they part company. In the clogging ensemble the contact degree collapses inside the throat and there are essentially no capsules downstream of it: the narrow gap admits at most a single capsule, and the rest of the suspension is held back by the arch. In the flowing ensemble the contact degree stays elevated through the throat and

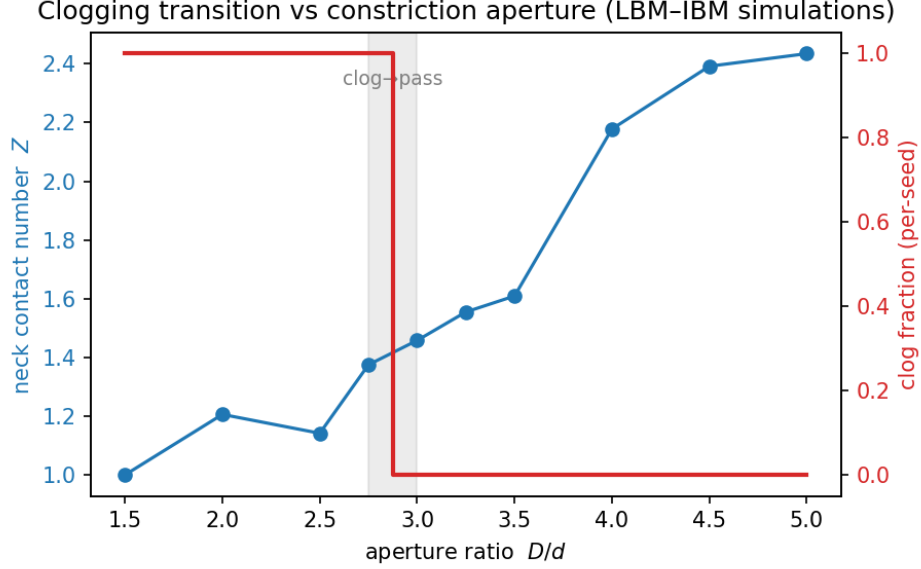


Figure 2. Clogging transition versus constriction aperture from the LBM-IBM runs. The neck contact number Z (blue) grows with the aperture ratio D/d , while the run-level clog fraction (red) drops from one to zero between $D/d = 2.75$ and 3.0 (shaded), consistent with the $D/d \approx 2-3$ arching threshold of the suspension-clogging literature.

the profile extends well past it, because capsules are continually carried through and populate the downstream channel. This streamwise asymmetry — a sharp loss of contacts and of downstream occupancy at the neck — is the structural fingerprint that the classifier exploits, and it is made explicit to the network through the node positions and the edge gap and approach-rate features.

4 Discussion

The central finding is that a single static snapshot of the contact network is enough to predict, with useful and transferable accuracy, whether a soft-particle suspension is clogging. The network never sees the dynamics; it is given only who is touching whom and the local geometry of those contacts, and from that it recovers the collective outcome. This supports a view of clogging as a property encoded in the instantaneous force-bearing network — the same picture that underlies the statistical-mechanical description of jamming — and it shows that the encoding is learnable directly from configurations rather than having to be summarised by hand-chosen order parameters.

Nothing in the method is specific to our capsules or to this geometry. The network is handed only a generic contact graph — particles as nodes, contacts as edges, with a few local geometric features — so the same recipe should carry over with little change to other particulate systems in which a collective state is written into the network of contacts. Granular flow in silos and hoppers, colloidal fouling of membranes and packed beds, and cell or capsule suspensions in confined vessels are all natural candidates: in each case a structural snapshot could be used to flag an incipient arch or jam, and more generally to detect emergent network signatures that are awkward to capture with a single hand-built order parameter.

The leave-one-run-out result is what gives the claim its weight. A random split over frames would have reported a flattering number, because frames from one simulation are highly correlated and a model can recognise a run it has partly seen. By removing an entire aperture, the LORO test asks the network to label configurations that are genuinely new, and the 0.87 mean shows that the contact-network signature it has learned is not run-specific. The single weak fold at $D/d = 2.75$ is instructive rather than disappointing: it is the aperture exactly on the transition,

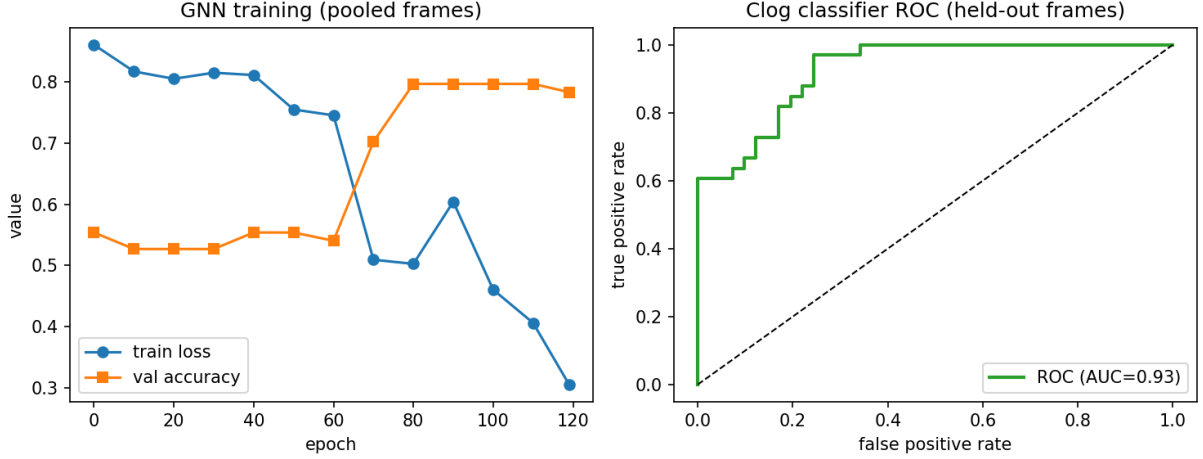


Figure 3. Graph-classifier training and discrimination. **Left:** training loss and validation accuracy over 120 epochs on the pooled frames. **Right:** ROC curve on held-out frames, with area under the curve 0.93; the dashed line is the chance diagonal.

and once it is removed the boundary is unrepresented in training. A classifier cannot be expected to place a sharp boundary it has never observed, and the human-meaningful conclusion is the same one the physics gives — the outcome right at the transition is intrinsically uncertain.

We also report a negative result that we think is useful. To strengthen the boundary we tried augmenting the data with a second, jittered realisation of each aperture, reasoning that a near-twin of the held-out boundary run in the training set would help. It did the opposite: the leave-one-run-out accuracy fell from 0.87 to 0.70. Near a sharp transition the outcome is so sensitive to the microscopic arrangement that two jittered realisations of the same nominal aperture can land on opposite sides of the boundary, so the augmentation injected label noise exactly where the model is most fragile. The lesson is that the binary run-level label is the limiting resource near the transition, and that more realisations do not help unless the label itself is made smoother; a continuous target such as the escaped fraction, or a per-frame “bridged” label, would be the more promising route.

Several limitations bound the scope of these results. The model is two-dimensional, so the arches are line-spanning rather than the three-dimensional vaults of a physical silo or vessel; we expect the qualitative contact-network signature to carry over, but the threshold and the absolute accuracies will shift in three dimensions. We fixed the capillary number, so deformability enters only through a single value and not as a control variable; sweeping it would test whether the same network generalises across softness as well as across geometry. The clogging label is binary and assigned at the run level, which, as the augmentation experiment showed, is the weakest point of the pipeline near the transition. Finally, the data set is modest — ten runs and a few hundred graphs — so the absolute numbers should be read as a proof of principle rather than a converged benchmark. None of these caveats undercuts the central result; they map out exactly what a larger study would need to vary.

5 Conclusion

We have shown that the clogging of a deformable-particle suspension at a microfluidic constriction can be predicted from the instantaneous contact network by an edge-aware graph neural network. A two-dimensional lattice-Boltzmann immersed-boundary model, run across ten aperture ratios, reproduces the expected arching threshold near $D/d \approx 3$; converting each frame to a graph of capsules and contacts and training a message-passing classifier yields an area under the ROC curve of 0.93 on held-out frames and, under a stringent leave-one-run-out test, a mean accuracy of 0.87

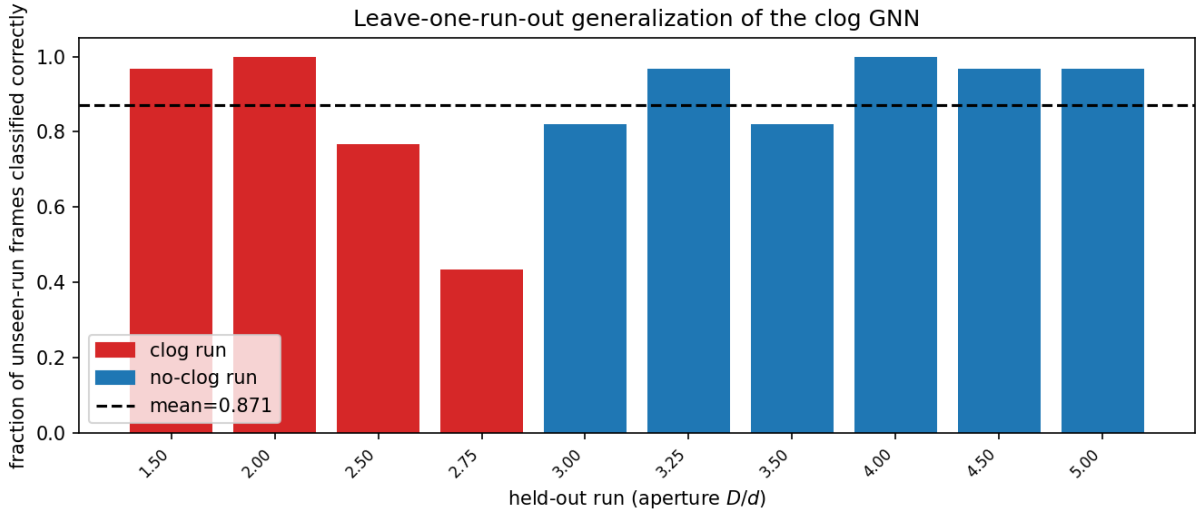


Figure 4. Leave-one-run-out generalisation. Each bar is the fraction of a held-out aperture’s frames classified correctly after retraining on the other nine apertures; red bars are clogging runs, blue bars flowing runs, and the dashed line is the mean (0.87). The model is strong away from the transition and weak only at $D/d = 2.75$, the aperture sitting on the clog/flow boundary.

on apertures the network has never seen, failing only at the single aperture on the transition. The instantaneous contact network thus carries a learnable and transferable signature of imminent clogging, and the work supplies a reproducible route from a mesoscale physics solver to a graph-based predictor. Natural extensions are to make the label continuous, to add the capillary number as a second control parameter, to move to three dimensions, and to use the trained network’s attributions to read off which contact-network features mark an incipient arch — turning the predictor into a tool for discovery as well as classification.

Data and code availability

The simulation engine, the graph-extraction and graph-neural-network code, and the scripts that generate the data set, train the classifier, run the leave-one-run-out evaluation and produce every figure in this paper are available from the author on reasonable request. The bulk trajectory output is regenerable from the provided scripts and fixed random seeds.

Declaration of competing interest

The author declares no competing interests.

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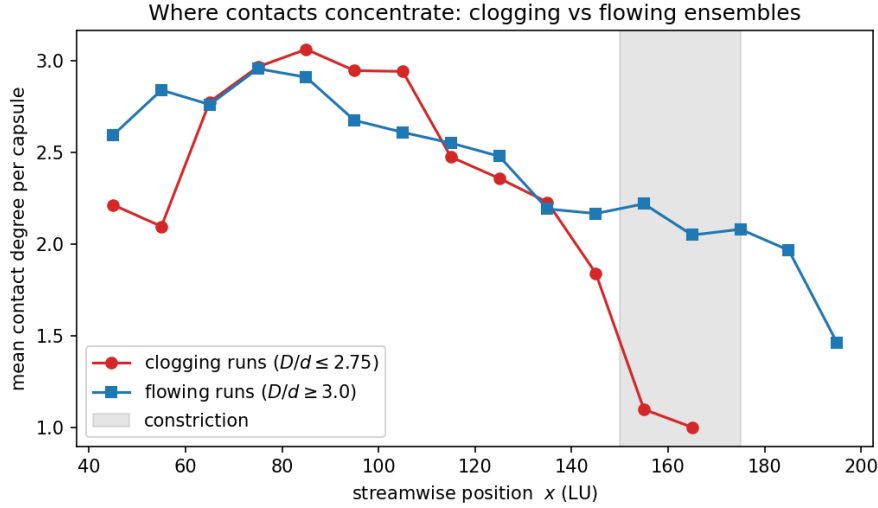


Figure 5. Mean contact degree per capsule as a function of streamwise position, averaged over all frames of the clogging runs ($D/d \leq 2.75$, red) and the flowing runs ($D/d \geq 3.0$, blue); the shaded band is the constriction. The two ensembles coincide upstream but diverge at the neck: the clogging ensemble loses its contacts in the throat and has no downstream occupancy (the arch admits at most one capsule), whereas the flowing ensemble stays connected through the neck and extends downstream.

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